Deep Learning
Final Project
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Final Project

The objective of the final project is to apply what you have learned in this course to a real world problem – a problem in your area of interest. You can apply any deep learning network to the problem, and you can use either TensorFlow or PyTorch for training and inference.

Potential Topics

Here are some potential topics that you can choose for your final project. (This list is not complete.)

- **Computer Vision** (Image classification, object detection, image segmentation, image generation, style transfer, super-resolution, facial recognition, pose estimation, emotion recognition, ...)
- **Sequence Processing** (Speech recognition, speaker identification, audio classification, DNA analysis, sound event detection, stock prediction, music generation, text to speech, source separation, ...)
- **Natural Language Processing** (Text classification, machine translation, summarization, question answering, information extraction, speech recognition, text generation, sentiment analysis, ...)
- **Tabular Data** (Fraud detection, disease diagnosis, equipment failure detection, recommender systems, imputation, multi-task learning, dimensionality reduction, clustering, sales forecasting, ...)

Here are some websites for trending research and datasets.

Trending Research
https://paperswithcode.com/

Dataset Search
https://datasetsearch.research.google.com/

1 Deliverables

1.1 Proposal

After you have selected a topic, a network, and a data set, submit a proposal of what you plan to do for the project. The proposal should be a few hundred words, and should address the following items.

- What problem did you select and why did you select it?
- What database/dataset will you use? Is it large enough to train a deep network?
- What deep network will you use? Will it be a standard form of the network, or will you have to customize it?
- What framework will you use to implement the network? Why?
- What reference materials will you use to obtain sufficient background on applying the chosen network to the specific problem that you selected?
- How will you judge the performance of the network? What metrics will you use?
- Provide a rough schedule for completing the project.

1.2 Presentation

You will give a 15 to 20 minute presentation (including 5 minutes to respond to questions) of your final project during the regular class periods on April 28 or April 30 (and, if necessary, during the Final Exam time on May 5 from 10:00-11:50am).

1.3 Final Report

The final report will include the following sections.

1. Introduction. An overview of the project and an outline of the report.
2. Project schedule. Provide an overview of your project schedule, including dates when tasks were started and completed. Use the project management features of GitHub to provide charts.
3. Description of the data set.
4. Description of the deep learning network and training algorithm. Provide some background information on the development of the algorithm and include necessary equations and figures.
5. Experimental setup. Describe how you used the data to train and test the network. Explain how you implemented the network in the chosen framework and how you judged the performance. How did you determine batch size? How did you determine training parameters (e.g., learning rate)? How did you detect/prevent overfitting, extrapolation, data imbalance? What type of data augmentation did you use? Describe your data loader.

6. Results. Describe the results of your experiments, using figures and tables wherever possible. Include all results (including all figures and tables) in the main body of the report, not in appendices. Provide an explanation of each figure and table that you include. Your discussions in this section will be the most important part of the report.

7. Summary and conclusions. Summarize the results you obtained, explain what you have learned, and suggest improvements that could be made in the future.

8. References. In addition to references used for background information or for the written portion, you should provide the links to the websites or github repos you borrowed code from.

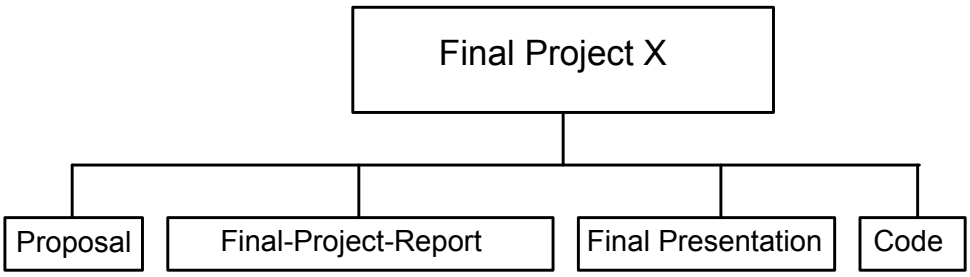
9. A separate appendix should contain documented computer listings (code).

2 Method for Delivery

You will use a GitHub repository for all of your code and reports. You will share your repository with me, so that I can clone the repository and download reports and/or code and follow your progress. You will use the **project management** (`https://docs.github.com/en/issues/planning-and-tracking-with-projects`) features of GitHub to plan your project and share milestones. The following describes how you will set up your GitHub repository.

- Make the following folders: Proposal, Final-Project-Report, Final-Presentation, Code.

- Create a GitHub repository for the project and name it Final-Project-X where `X` is your name. Then, push the folders described in the previous item into the repository that you created. You should have a markdown file (`README.md`) that explains the structure of the repoistory and how it works. Make it as clear as possible. The GitHub repository should have the following structure.



- Write your final proposal and save it as a PDF file. Place the proposal in the Proposal Folder and push it to your GitHub repo. At this time you will post your GitHub repo link to canvas so that I can clone your repo. (March 26)

- Save all of your code for the final project in the Code folder and push them regularly to your GitHub repo. For example, if you write different code for differnt parts of the project, give them descriptive names and write a README file that describes how to use them. This README file should explain what order the code needs to be run in (e.g., code to fetch data should be run first, then code to preprocess, and then modeling, etc.) and a short description of what each script does.

- Create a presentation document (for example, using powerpoint) for your presentation. Save the presentation as a PDF file and move it to the Final-Presentation folder and push it to your GitHub repo. (April 28)
- Write your final project report and save it as a PDF file. Move it to the Final-Project-Report folder and push it to your GitHub repo. (May 5)

3 Grading

The final project will be worth 25% of your course grade. The final project grade will be made up of two equal parts.

- The first half of the final project grade will come from your technical reports and working code.
- The second half of the final project grade will come from your presentation and your answers to oral questions following the presentation.
- Online students who are available during class time will give their oral presentations to the class through zoom link. For online students who are not available during class time, another suitable time will be arranged for a live zoom presentation.

4 Full Schedule

Date	**Item**
03/11/26	Project description in Canvas
03/26/26	Proposal Due (GitHub link posted to Canvas.)
04/27/26	Oral presentation slides (PDF) posted in GitHub
04/28/26	Oral presentations – First Day
04/30/26	Oral presentations – Second Day
05/05/26	Oral presentations – Third Day (if needed)
05/05/26	Final reports due